

New Opportunities in Optical Networks Enabled by Hollow-Core Fibres

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Abstract *Hollow-core fibres enable performance and applications that are challenging to achieve in conventional solid-core fibre systems. Here we review demonstrations in intensity modulation direct-detection transmission and simultaneous power-and-data delivery that offer opportunities for new services and enhanced performance in optical communication networks. ©2026 The Author(s)*

Background

Although hollow-core fibres (HCFs) have been the subject of research for over two decades [1], recent advances in their performance have generated significant excitement in the fibre optics and communications community [2]. Light guidance in air rather than glass, leads to some unique characteristics that have long motivated research in this area. Firstly, hollow-core guidance results in lower transmission latency, since light is no longer slowed down by propagation in the relatively dense glass core. This corresponds to an approximately 30% reduction in latency relative to standard single-mode fibres; over a distance of 5 km, the gain in arrival time can be as large as 5 μ s. In addition, optical nonlinear effects that dominate system design in fibre communications, are suppressed by as much as three orders of magnitude when light is guided in a hollow core, offering the potential to rethink both system architectures and signal processing approaches.

Early research on HCFs focused on photonic bandgap fibre designs, where a carefully fabricated periodic lattice of air holes surrounds the hollow core. Considerable effort from several research groups led to significant progress in both the propagation loss and the optical bandwidth supported by these fibre structures [1, 3]. Nevertheless, the propagation loss remained almost an order of magnitude higher than in standard fibres, which limited early applications mainly to specialised use cases.

The introduction of a new guiding mechanism based on anti-resonant designs implemented in negative-curvature cladding structures offered new opportunities [2, 4, 5]. Crucially, this approach enables HCF designs that avoid glass nodes in the microstructure, which have typically

been associated with increased propagation loss. This has proven to be the key to reducing HCF loss to levels that reach, and even surpass those of conventional single-mode fibres, motivating rapid industrial uptake and giving rise to the intense research activity witnessed today.

One of the attractions of microstructure fibre technology, and HCFs in particular, is the high degree of fibre customisability. This allows for example, HCFs to be engineered for operation across a wide range of wavelengths, opening up fibre optic applications across the spectrum. However, low-loss anti-resonant HCFs generally feature relatively large core dimensions, necessitating the development of careful interfacing strategies with standard single-mode fibre systems. Equally important is the development of fibre optic devices based on HCFs, to minimise propagation in solid-core fibres within HCF-based systems, and significant progress in this direction has already been achieved [6].

This paper reviews recent research that exploits the potential offered by HCFs to realise implementations in communication systems that would be particularly challenging to achieve in solid-core single-mode fibre. Applications in simplified intensity modulation direct-detection (IM/DD) transmission and simultaneous power-and-data delivery are discussed.

Prospects for High-Speed IM/DD Transmission

The low latency and high linearity of HCF, especially when combined with the low loss and broad bandwidth offered by nested anti-resonant structures, create an attractive environment for optical transmission. These characteristics are combined with low dispersion across the propagation

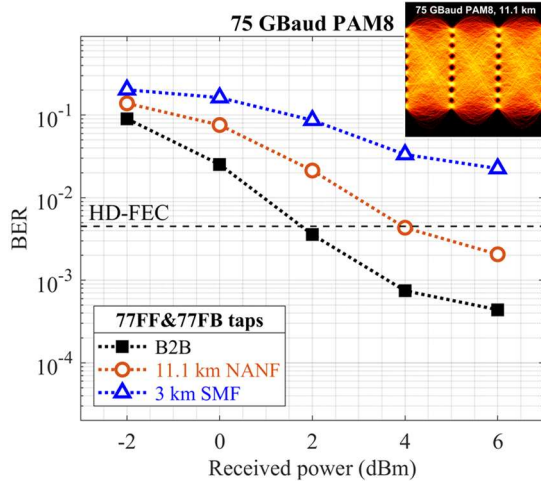


Fig. 1: Comparison of the transmission performance of a 225-Gbit/s PAM8 signal in 11.1 km HCF versus 3 km of standard single-mode fibre. (Adapted from [8])

window of the fibre. While such properties primarily offer increased tolerance and implementation simplicity in coherent systems, they are much more consequential for IM/DD transmission, where dispersion-induced power fading can directly and severely limit the available signal bandwidth. While the bandwidth of a standard single-mode fibre may be limited to approximately 20 GHz at 1550 nm over a distance of 5 km, the corresponding bandwidth in HCF can be more than three times as broad, and vary less with wavelength [7]. These characteristics of HCF, combined with its low nonlinearity, open a route to extremely high bit rate transmission whilst retaining the simplicity of IM/DD modulation formats.

Recent experiments tested implementations for the transmission of 225 Gbit/s IM/DD signals on a single wavelength over 11 km of HCF [8]. Different pulse-amplitude modulation (PAM) orders, namely PAM-4, PAM-6 and PAM-8, were considered to explore the trade-off between dispersion tolerance for signals with a varying analogue bandwidth and the signal-to-noise ratio requirements associated with high-order PAM detection. Transmission over this distance in HCF was compared favourably against transmission over just 3 km of standard single-mode fibre (Fig. 1) [8].

Simultaneous Power-and-Data Delivery

The availability of a highly linear waveguiding medium opens opportunities to explore service implementations that would be either unfeasible or impractical with incumbent infrastructure. Power delivery over the telecommunications cable is a representative example. This capability has been effectively lost with the removal of copper while transitioning to all-fibre networks. Restoring the ability to power devices through the network

enhances resilience, enables operation even at remote locations, and creates opportunities for new service concepts. Hollow-core fibres are well-suited to this application [9], since apart from their intrinsically high linearity, light propagation in air also enables safe coupling of high-power beams offering a substantially increased damage threshold. Moreover, the low loss of nested anti-resonant HCFs allows power signals to be transmitted over long distances with little attenuation, thereby improving the overall system efficiency relative to solid-core fibre systems, while also offering the flexibility to engineer the operating wavelength independently of the constraints imposed by material absorption in solid-core fibres.

To demonstrate this potential, we experimentally evaluated the simultaneous transmission of a high-power optical beam, generated from filtered and amplified ASE from an erbium-doped fibre amplifier, together with high bit-rate WDM data signals over 3-km lengths of fibre [10]. In standard single-mode fibre, the data channels were severely impaired due to the power signal. In contrast, propagation in HCF showed no observable performance degradation in the presence of the power signal.

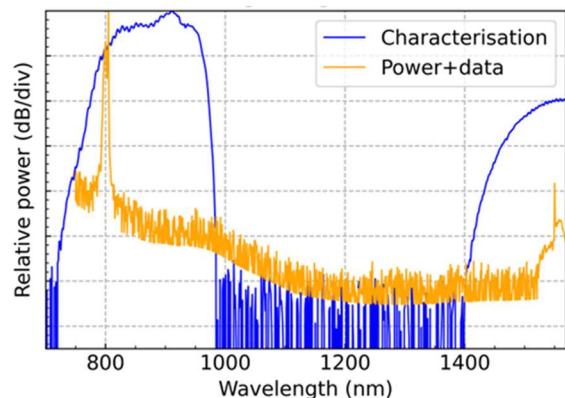


Fig. 2: Two guiding windows within the same HCF allow simultaneous coupling and propagation of a power signal at 800 nm and a data signal at 1550 nm. (Adapted from [10])

An additional possibility is to exploit multiple transmission windows within the same HCF. This concept was demonstrated experimentally using a fibre designed for low-loss operation around 800 nm. The fibre exhibited an additional transmission window supporting propagation at telecommunication wavelengths. Since laser power converters are commonly available and more efficient at shorter wavelengths, power transmission was carried out at the primary transmission window of this fibre at 800 nm. At the same time, independent data transmission was implemented at 1550 nm [10].

Conclusions

The emergence of hollow-core fibre technology has enabled systems offering performance that is difficult to achieve with conventional solid-core fibres. We reviewed use cases in optical communications in which HCFs were used to either add functionality or offer scalability to the network in a sustainable fashion. With the wider deployment of HCFs, the technology has the potential to support the emergence of multi-service optical networks, while delivering improved energy efficiency relative to solid-core solutions.

Acknowledgements

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